

SESSION 1

From Engineering or Disciplinary Problem to Machine Learning Problem

Applied AI Design Workshop

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Goal: leave with a defensible ML problem statement for a research, teaching, or operational challenge.

Session focus

- Prediction objective
- Inputs and target
- Unit of observation
- Baseline model
- Success criteria
- Data-readiness risks

Workshop outcomes

By the end of the Day 3 foundations block, faculty should be able to design a credible starting point for applied AI — before selecting a tool or model.

Translate
Turn disciplinary
questions into ML
tasks

Differentiate
Regression,
classification,
clustering,
forecasting

Baseline
Establish a simple
reference model

Evaluate
Choose initial
metrics based on
consequence of
error

Connect
Position classic
ML within applied
AI workflows

Today's emphasis: problem framing is a design activity, not just a coding step.

Day 3 schedule and flow

From engineering problem to ML problem

Now

Classic ML overview

Framing

Invited presentation: From Prompts to Agents

Context

Break

Regression fundamentals

Classification fundamentals

Methods

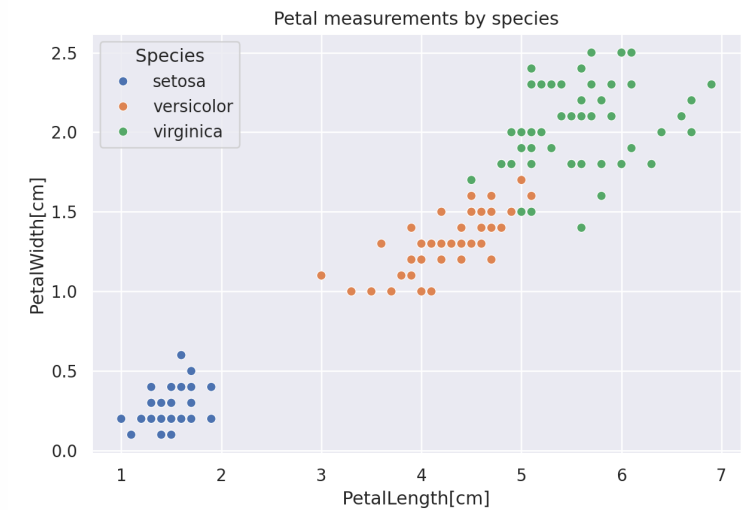
Hands-on baseline model activity

Practice

Embedded Technical Skills

pandas → visualization → regression → classification → complete workflow

Session	Tool skill	What faculty practice
Session 1	pandas data loading	Load CSV, inspect rows/columns, define inputs and target
Session 2	visualization	Scatterplot, histogram, target count plot
Session 3	regression in scikit-learn	Split, baseline, train, predict, MAE and R^2
Session 4	classification in scikit-learn	Majority baseline, logistic regression, confusion matrix, precision/recall/F1
Session 5	complete workflow	Load → inspect → visualize → split → baseline → train → evaluate



```
df = pd.read_csv("iris.csv")
X = df[["PetalLength_cm", "PetalWidth_cm"]]
y = df["Species"]
model.fit(X_train, y_train)
```

Learning activities for Session 1



Participant output: one polished ML problem statement ready to reuse in course planning, research ideation, or proposal development.

The core design question

Before asking “which model?” ask:

What decision, prediction, or insight should improve because this model exists?

Prediction

Estimate an unknown or future value

Decision support

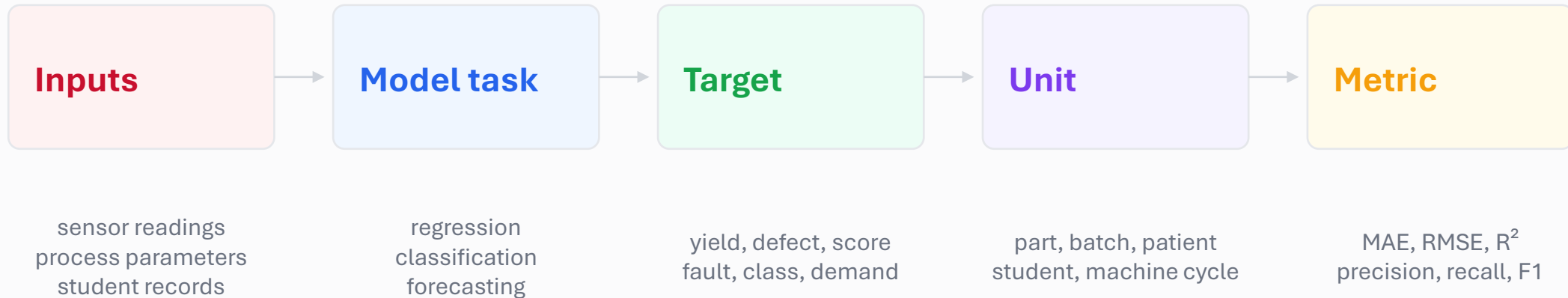
Rank, flag, recommend, or screen

Discovery

Reveal patterns for further study

The problem framing determines the data, model family, metric, and interpretation.

Anatomy of an ML problem statement



Using _____ as inputs, develop a _____ model to predict or identify _____ for each _____, evaluated by _____.

Define the four essentials

Input variables

What the model is allowed to know

Temperature, pressure, vibration, assignment activity

Target variable

What the model is trying to estimate

Yield, final strength, fault status, retention risk

Unit of observation

One row or prediction case

Part, batch, machine cycle, patient, student, building-day

Output format

How the model returns information

Continuous value, probability, class label, cluster ID

Rule of thumb: inputs must be available at the time the prediction is made.

Choose the ML task from the question

Regression

How much?

final strength, energy, yield

Classification

Which category?

defect / no defect, pass / fail

Clustering

What groups exist?

operating regimes, student profiles

Forecasting

What happens next?

demand, sensor trajectory

Anomaly Detection

Is this unusual?

faults, outliers, rare events

Examples across disciplines

Domain	Question	ML task	Possible target
Manufacturing	Will this run produce an acceptable part?	Classification	Accept / defect
Materials	What final strength will result?	Regression	Tensile strength
Energy	What will demand be tomorrow?	Forecasting	kWh demand
Education	Which learners need support?	Classification	Support-risk category
Bioprocessing	Which runs have similar behavior?	Clustering	Operating group

Same dataset, different question → different model design.

Baselines and success criteria

A baseline answers: “What would happen if we did something simple?”

Regression baseline
Predict the mean or median
of the training target

Classification baseline
Predict the majority class or
simple rule

Operational baseline
Use current practice, expert
rule, or manual workflow

Success criteria combine technical performance and practical consequence:

Metric threshold: e.g., recall $\geq$ 85% for rare faults

Practical constraint: e.g., fewer than 5 false alarms per week

Scientific usefulness: e.g., interpretable features for follow-up study

Adoption criterion: e.g., model informs a real decision in the workflow

Data-readiness checks before modeling

Labels

Is the target measured reliably?

Timing

Are inputs available before prediction?

Leakage

Did we accidentally include the answer?

Sample size

Are there enough observations for the task?

Representation

Do data cover relevant conditions?

Error cost

What errors matter most?

Data quality is not a preprocessing detail; it determines whether the ML problem is answerable.

problem-framing canvas

Disciplinary problem

What challenge or opportunity motivates AI?

Prediction objective

What should be predicted, flagged, ranked, or discovered?

Unit of observation

What does one row or prediction case represent?

Inputs

What data are available before the prediction?

Target

What is the outcome or label?

Baseline + metric

What simple approach and evaluation metric will you compare against?

What would make your problem statement scientifically or operationally defensible?

Worked example: machine-condition prediction

Problem

Reduce unexpected downtime
in a manufacturing cell

Objective

Predict fault risk in the next
24 hours

Task

Binary classification

Inputs

current, vibration,
temperature, cycle time

Unit

one machine cycle or shift

Target

fault / no fault label

Baseline

predict no fault

Metric

recall for fault class

Discussion prompts

What decision would change if the model worked?

Can the target be measured reliably?

Are inputs available before the prediction?

What type of error is most costly?

What simple baseline must be beaten?

Would strong prediction imply understanding or causality?

Use these prompts to move participants from “I have data” to “I have a well-posed ML design problem.”

Example Machine Learning Template

Write your ML problem statement:

Using _____ as inputs, develop a _____ model to predict or identify _____ for each _____, evaluated by _____.

What is the decision?

What data are available?

What is the target?

What is the baseline?

Before you begin

You should have:

- 1 A problem statement
- 2 A selected task type
- 3 Defined inputs, target, unit of observation
- 4 A baseline model or current-practice comparison
- 5 An initial metric and practical success criterion
- 6 One risk to resolve before modeling

Transition: Session 2 will place this framed problem into the broader classic ML landscape.

Session 1 technical skill: load, inspect, frame

Goal: make inputs, target, units, task, baseline, and metric explicit before modeling.

```
import pandas as pd

df = pd.read_csv("iris_workshop.csv")
df.head()
df.shape
df.dtypes
df.isna().sum()
```

What faculty should identify

- What does one row represent?
- Which columns are inputs?
- What is the target?
- What are the units?
- Is the task regression, classification, clustering, or forecasting?

Session 1 notebook

01_session1_loading_and_framing.ipynb

Instructor prompt

“Before we choose a model, describe the problem using the table: unit of observation, inputs, input units, target, task, baseline, metric.”

Session 1 example: the Iris problem as a framing template

Use a familiar dataset to model the vocabulary faculty will apply to their own disciplines.

Unit

One flower

Inputs

Petal length, petal width

Units

Centimeters

Target

Species

Task

Multiclass classification

Metric

Accuracy

```
input_columns = ["PetalLength_cm", "PetalWidth_cm"]  
target_column = "Species"
```

```
X = df[input_columns]  
y = df[target_column]
```

Key takeaways

Applied AI design begins with problem formulation, not model selection.

Define the prediction objective in decision-relevant language.

Separate inputs, target, output, and unit of observation.

Select the ML task based on the question being asked.

Use baselines and metrics to avoid vague claims of “good performance.”

Check data availability, leakage, labels, and error consequences early.

Next: Classic machine learning overview — matching framed problems to model families.